# Time Is the Baseline to Beat: Lead Time over a Timer as the Test for VLA Failure Monitors

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***Abstract*—Runtime failure monitors for vision-language-action (VLA) policies are often evaluated under success-terminated, time-limited protocols, where a failed episode is one that reaches its time limit. There, elapsed time is a strong failure signal: on RoboCasa365, a timer calibrated at a 5% false-positive rate (FPR) flags every failure, and a model that sees only elapsed time separates failures as well as one that reads the policy. Without a duration-only comparator, even timeliness metrics cannot show what a monitor adds. We measure how much earlier than a per-task timer a monitor alarms at the same target FPR, and fuse the two by an OR rule with a split-conformal FPR bound. On 760 held-out rollouts of one VLA on 38 tasks, with hypotheses fixed in advance, the SAFE failure score adds 0.030 of the time limit to the timer and the sequence detector Hide-and-Seek, with three seeds averaged, 0.071 (95% CI [0.022, 0.157]); at matched false alarms they shrink to 0.015 and 0.056, with CIs including zero. Input ablations that see only elapsed time or another rollout's features add at most 0.012. On unseen tasks, where no per-task timer exists, Hide-and-Seek still adds 0.127 over a global timer, SAFE about as much as random noise. The ordering holds on recorded rollouts of a second VLA, and on FIPER's benchmark the test credits FIPER. Stage 2 tests the second VLA on new rollouts and whether earlier alarms raise success through retry.**

## I. Introduction

Vision-language-action (VLA) policies [11, 4] fail often enough that a deployed robot must notice when an attempt is going wrong. Runtime monitors map the policy's hidden state to a failure score [7], or score how unusual its embeddings or actions are relative to successful rollouts [29, 23]. They are compared with each other on ROC-AUC, balanced accuracy and, increasingly, timeliness [23, 21], but rarely with what the robot knows for free: how long the attempt has been running.

Under success-terminated, time-limited protocols such as those of RoboCasa365 [19] and LIBERO [14], an episode ends at the first success or at a fixed time limit, so a failure is, by construction, an episode that ran out of time. In RoboCasa365 every one of our failed rollouts ran to its limit, and a timer at 90% of the limit flags all of them at 4.7% FPR. Elapsed time also explains separation: an LSTM trained on zero inputs, which sees only elapsed time, matches the ROC-AUC of the same LSTM trained on the policy's features (0.888 against 0.878). On such benchmarks, a monitor can add only one thing over a timer: an earlier alarm at the same false-alarm rate (Fig. 1), whose value Stage 2 tests by acting on it.

We therefore evaluate monitors by their *warning lead* over a per-task timer that alarms once an attempt has run longer than the task's successful attempts usually take. We hypothesize that the policy's internal state carries warning that elapsed time does not: fused with this timer, a learned failure score should raise the mean lead at the same FPR, and the gain should vanish when the score cannot see the rollout. Our contributions are:

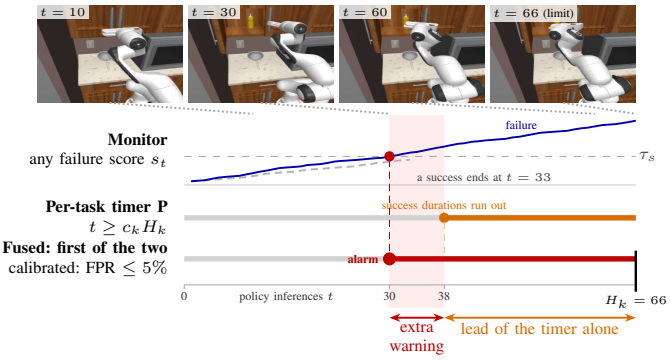



Fig. 1. In success-terminated, time-limited benchmarks every failed rollout runs to its time limit $H_k$, so a per-task timer that alarms once an attempt outlasts the task's successes flags all failures. A monitor helps only if it alarms *earlier*. We fuse any monitor with the timer (first alarm wins), calibrate both for a 5% false-alarm rate, and measure the extra warning over the timer. Example: a failed XR-1 *OpenCabinet* rollout, monitor score from SAFE.

- **A duration-only comparator for timeliness metrics:** warning lead over a timer at matched FPR, on known and unseen tasks, with input ablations that separate rollout-specific evidence from learned elapsed time (Sec. III, IV).
- **Time-aware fusion:** an OR rule that fuses any monitor with a timer, whose split-conformal calibration bounds the FPR in finite samples (Proposition 1).
- **An evaluation on 38 RoboCasa365 tasks** with hypotheses fixed before the held-out test. SAFE adds 0.030 of the time limit over the per-task timer and Hide-and-Seek [21] 0.071, a warning that develops during the rollout (Sec. VI). On XR-1's unseen tasks, only Hide-and-Seek's gain carries over; the ordering holds on a second VLA, and on FIPER's benchmark the test credits FIPER.

Most failures we annotated had reached no irreversible state by the limit, which is why lead, not recall, is the quantity to report.

## II. Related Work

**Failure detection from policy internals.** SAFE learns a failure score from a VLA's last-layer features and thresholds it with a functional conformal band [7]; we reuse its implementation. Follow-ups change the training data [22], labels [21, 17],

or policy [25]. None reports an elapsed-time detector. SAFE identifies the duration shortcut and removes it by evaluating rollouts at equal length (its App. B.5) and, on its real robot, by running every rollout to a fixed horizon [7]. These controls suit fixed-horizon evaluation; under success-terminated execution, duration is a shortcut [6] for offline metrics but legitimate online evidence, much as time limits make the remaining time part of the agent's state in reinforcement learning [20].

**Action-space and success-only detectors.** STAC [1] and VLA-FAIL [24] compare overlapping action chunks, and ActProbe [9] feeds chunk consistency and magnitude to a task-conditioned LSTM, beating internal-feature baselines on RoboCasa. FAIL-Detect [29] and FIPER [23] score inputs, outputs or embeddings against successful rollouts only. FIPER alarms when two conformal scores *both* exceed their thresholds, which suppresses false alarms; we alarm when *either* crosses, as the two signals catch failures at different times. CheckVLA [16], the closest work on our benchmark, verifies each action chunk with a world model on RoboCasa365 at a conformal 5% rate of unnecessary interventions and repairs it with the same VLA. Neither FIPER nor CheckVLA compares against elapsed time.

**Time, calibration and intervention.** FIPER argues that late detection is not enough [23] and Hide-and-Seek weights accuracy by time [21]. Our lead is one minus FIPER's detection time and, when failures run to the limit, determines Hide-and-Seek's TWA at equal FPR (Sec. VI-A): the metric is not new, the duration-only comparator is. Lead at a fixed false-alarm budget is the accuracy–earliness trade-off of early time-series classification [28]. Split-conformal thresholds give finite-sample false-positive control under exchangeability [3]; we extend the construction to a two-signal OR rule (Sec. IV-D). RESCUE [2] finds that good detection scores do not guarantee lower intervention cost, and CheckVLA couples detection to repair; an earlier alarm matters only if acting on it raises success.

## III. Problem Setup

**Time.** The simulator runs *control steps* at RoboCasa's default 20 Hz. The policy executes each 16-step action chunk in full, so *policy inferences* $t = 1, 2, \ldots$ (0.8 s each) are our clock, and a monitor decides only at an inference. Task $k$'s limit of $T_k$ control steps is $H_k = \lceil T_k/16 \rceil$ inferences (29 to 272, or 22.5 to 217.5 s). Every fraction of time ($t/H_k$, $c$, $\ell$) is a fraction of $H_k$.

**Monitoring.** A monitor watches a rollout of a known task $k$ and decides at each inference $t$ whether to alarm, using only the policy features so far, $t$, the task and $H_k$. The first alarm is at $T$ (in inferences). FPR is the fraction of successful rollouts with any alarm. A rollout of length $L$ ends at success or at $H_k$. For a failure ($L = H_k$) the *warning lead* is $\ell = (L - T)/H_k$, one minus the normalized detection time of [23], with $\ell = 0$ for a missed failure; an alarm at the final inference counts with $\ell = 0$. The mean lead $\bar{\ell}$ over failures combines recall and earliness. Detectors are compared at thresholds set on a calibration set for FPR $\leq \alpha = 0.05$.

## IV. Method

### A. Monitor scores

**Policy score.** At each inference we take the final-layer action-token hidden state of the policy's action head (1,024 dimensions). The SAFE MLP [7] maps it to a failure score $u_t^{(j)}$ for each of three training seeds $j$. With per-task location $m_k^{(j)}$ and scale $\sigma_k^{(j)}$ estimated on training rollouts, the policy score is

$$s_t = \frac{1}{3} \sum_{j=1}^{3} \frac{\max_{t' \leq t} u_{t'}^{(j)} - m_k^{(j)}}{\sigma_k^{(j)}}. \tag{1}$$

The running maximum makes $s_t$ monotone, so it does not change once a threshold is crossed. On unseen tasks, $m^{(j)}$ and $\sigma^{(j)}$ are pooled over training tasks. Success-only alternatives (Mahalanobis [13], kNN [26], PCA-KMeans [15] and RND [5, 8]) are fitted on training successes and aggregated the same way.

**Sequence score and input ablations.** Hide-and-Seek [21] trains an LSTM on a 128-dimensional PCA of the same hidden state with rollout labels only: a margin loss pushes each failure's peak score above every success's, and an onset loss acts within failures. A recurrent model can learn how long each task takes, so we retrain it unchanged on ablated inputs: *zero input*, so it sees only elapsed time; *shuffle*, a random same-task rollout's features at the same inference, so it also sees the task's typical trajectory; and *first step*, the rollout's own first-inference features, repeated, so it also sees the initial scene. ActProbe [9] reduces to chunk magnitude here, as our chunks never overlap.

### B. Time baselines

The *global timer* G alarms once $t \geq c\,H_k$, with one fraction $c \in (0, 1]$ for all tasks. The *per-task timer* P uses $c_k = \min(1, \rho_k + \delta)$, where $\rho_k$ is the 95th percentile of task $k$'s training success durations (as fractions of $H_k$) and the shared margin $\delta$ is set by Proposition 1. P needs successful rollouts of each task, but no failures and no policy features.

### C. OR-fusion

The fused monitor alarms at

$$T = \min\{\, t : s_t > \tau_s \text{ or } t \geq c\,H_k \,\}. \tag{2}$$

Neither signal is restricted in time: the score may alarm early, and, once calibration certifies a usable point (Sec. IV-D), the timer alarms on every failure by $c\,H_k$. Any lead gain over the timer comes from failures on which the score crosses first.

### D. Calibration with a finite-sample FPR guarantee

A successful rollout $i$ of length $L_i$ (in inferences) raises a false alarm under $(\tau_s, c)$ if and only if $M_i = \max_{t \leq L_i} s_t > \tau_s$ or $\rho_i = L_i/H_{k_i} \geq c$, so its false alarm depends only on the pair $(M_i, \rho_i)$. Fix in advance a path $\gamma(\lambda) = (\tau_s(\lambda), c(\lambda))$, $\lambda \geq 0$, with both coordinates nondecreasing and $c \leq 1$, so that a larger $\lambda$ is more conservative. For each calibration success define

$$V_i = \min\{\lambda : M_i \leq \tau_s(\lambda) \text{ and } \rho_i < c(\lambda)\}, \tag{3}$$

the least conservative point on the path at which rollout $i$ stays silent, and $V_i = \infty$ if none is: a success that lasts the whole limit ($\rho_i \geq 1$) alarms at every point. Because the path is monotone, rollout $i$ is silent at $\lambda$ exactly when $\lambda \geq V_i$.[1]

*Proposition 1:* Let $n$ calibration successes and a new successful rollout be exchangeable, and let the path $\gamma$ not depend on the calibration successes. Set $\hat{\lambda} = V_{(r)}$, the $r$-th smallest $V_i$ with $r = \lceil (n+1)(1-\alpha) \rceil$, or $\hat{\lambda} = \infty$ if $r > n$. If $\hat{\lambda} = \infty$, the calibration rule cannot certify a usable point at the target and the monitor is disabled: it raises no alarm. Otherwise it runs at $\gamma(\hat{\lambda})$. Then, over calibration draws, $\Pr(\text{monitor enabled and false alarm on the new success}) \leq \alpha$.

*Proof:* An enabled monitor alarms on the new success if and only if $V_{n+1} > \hat{\lambda}$, and a disabled one ($\hat{\lambda} = \infty$) never does, so the event is $\{V_{n+1} > V_{(r)}\}$ with $V_{(r)} = \infty$ for $r > n$. By exchangeability, the rank of $V_{n+1}$ among $V_1, \ldots, V_{n+1}$ is uniform (ties broken at random), so $\Pr(V_{n+1} > V_{(r)}) \leq 1 - r/(n+1) \leq \alpha$: the split-conformal quantile lemma [3]. ■

The path may use any data except calibration successes, such as development rollouts or calibration *failures*. The guarantee is per trajectory and marginal over tasks and calibration draws, not a cap on a finite test set or on each task (on the held-out set, one task had 4 of 10 successes alarmed); a per-task version needs 19 calibration successes per task.

**Choosing the path.** For every detector fused with P we use the path $\delta = \lambda$, $\tau_s = u + s\lambda$, with $(u, s)$ the grid point of largest mean lead on calibration failures, each point placed at its own $\hat{\lambda}$. This choice also uses the calibration successes, so Proposition 1 bounds each candidate path, not the chosen one; over 200 re-splits of the successes, the choice raised the expected test FPR from 0.047 (P) to about 0.055. A path chosen on other rollouts restores the bound: chosen on the validation set, Hide-and-Seek + P still gains 0.072 at 0.061 FPR (SAFE + P: 0.007), but both CIs then include zero (Hide-and-Seek: $[-0.069, 0.143]$). Stage 2 fixes every path in advance (Sec. VII).

## V. Hypotheses

**Early warning (H1).** At thresholds calibrated for FPR $\leq 0.05$, OR-fusing a policy-internal score with the per-task timer P increases the mean warning lead over P alone.

**Policy evidence (H2).** Scores that carry no information about the rollout (random noise, or the policy score resampled within the task) do not increase the lead when fused with P.

**Second policy (H3).** On a second VLA, RLDX-1 [10], SAFE + P and Hide-and-Seek + P also lead P, including at matched false alarms.

**Value of earlier warning (H4).** Within a budget of $1.25\,H_k$, resetting and retrying when Hide-and-Seek + P alarms yields a higher success rate than resetting when P alarms.

We fixed H1, H2, P and the lead metric before scoring the held-out test set, first for SAFE and then for Hide-and-Seek. Stage 2 addresses H3 and H4 (Sec. VII).

[1] $\lambda$ takes the finitely many values at which the timer (one inference) or the score changes the alarm, so the minimum is attained.

## VI. Experiments

### A. Setup

**Policy and tasks.** We use the released Xiaomi-Robotics-1 (XR-1) RoboCasa365 checkpoint [27] on the 38 official RoboCasa365 tasks [18, 19] whose published success lies between 5 and 45 of 50 (10 atomic and 28 composite tasks).

**Data.** Disjoint sets of rollouts serve one purpose each. A *training set* (1,824 rollouts) fits every learned detector and P's $\rho_k$. A *validation set* (380 successes and 380 failures, 10 per class per task), with its own calibration set (3 successes and 3 failures per task), was used to develop the timers and the fused detectors. A *held-out test set* of the same size, with new seeds, and a new calibration set of 380 successes and 380 failures served the tests of H1 and H2.

**Unseen tasks.** Each task is held out once (4 folds): monitors are retrained and thresholds calibrated without it, with OOD references and score normalization pooled over the other tasks. P needs the task's own successes, so the comparator is G. (Hide-and-Seek's evaluation [21] uses random 7/3 task splits instead.)

**Baselines.** We compare three families (Table I). *OOD detectors*, fitted on successful rollouts only: Mahalanobis distance [13], cosine k-NN [26], PCA-KMeans [15] and RND [8]. *Failure classifiers*, trained with rollout labels: SAFE-MLP [7], ActProbe [9] and Hide-and-Seek [21]. *Elapsed time*, with no policy access: G and P. Token-uncertainty baselines need discrete action tokens and STAC [1] overlapping chunks, which XR-1 does not produce.

**Metrics.** Following [23, 21], at thresholds calibrated for FPR $\leq 0.05$ we report test FPR, balanced accuracy (bACC; equal to weighted accuracy on our balanced sets), time-weighted accuracy $\mathrm{TWA} = \frac{1}{2}(\bar{\ell} + 1 - \mathrm{FPR})$ and detection time $\mathrm{DT} = 1 - \bar{\ell}$, a miss counting as 1. As in [21], learned detectors are trained with 3 random seeds and tables give the mean and std of single models. For H1 and H2, each detector averages the 3 seeds' scores; as the seed std ignores test-set variation, brackets give 95% bootstrap CIs over tasks, then rollouts, that also resample the calibration set and refit all thresholds [12]. *Matched* gains compare with P loosened on the test set to the fused detector's false alarms, which favors the timer.

### B. Standalone failure detection

Evaluated on their own, as in prior work (Table I), the OOD detectors and failure classifiers catch 36% to 69% of the failures on seen tasks at 5% FPR. The two timers, which never read the policy, catch all of them (bACC 0.97 and 0.99, against at most 0.82). The global timer alarms last, but P (DT 0.775) alarms earlier than every learned detector except Hide-and-Seek (0.754), which alone exceeds its TWA. On unseen tasks, Hide-and-Seek is the only detector above the global timer's TWA, and ActProbe's thresholds almost never fire.

### C. Warning lead over the per-task timer

**SAFE: a small gain (H1, H2).** On the held-out test set (Table II), SAFE + P leads P by 0.030 of the time limit ([0.007, 0.093]) at 0.061 FPR. Random and within-task resampled

TABLE I

STANDALONE FAILURE DETECTION WITH XR-1 (HELD-OUT TEST SET, 380 + 380 ROLLOUTS) AT THRESHOLDS CALIBRATED FOR 5% FPR. MEAN AND STD OVER 3 SEEDS; **BOLD**/UNDERLINE: BEST/SECOND BEST.

| | **Policy: XR-1 (Success Rate: 55.6%)** | | | | |
|---|---|---|---|---|---|
| | | Seen tasks | | Unseen tasks | |
| | Method | bACC↑ | TWA↑ | bACC↑ | TWA↑ |
| OOD | Mahalanobis | $0.818^{\pm 0.001}$ | $0.582^{\pm 0.001}$ | $0.726^{\pm 0.000}$ | $0.543^{\pm 0.000}$ |
| OOD | Cosine k-NN | $0.788^{\pm 0.000}$ | $0.577^{\pm 0.000}$ | $0.729^{\pm 0.000}$ | $0.544^{\pm 0.000}$ |
| OOD | PCA-KMeans | $0.796^{\pm 0.001}$ | $0.578^{\pm 0.001}$ | $0.728^{\pm 0.001}$ | $0.542^{\pm 0.001}$ |
| OOD | RND | $0.793^{\pm 0.002}$ | $0.572^{\pm 0.001}$ | $0.728^{\pm 0.001}$ | $0.547^{\pm 0.001}$ |
| Class. | SAFE-MLP | $0.763^{\pm 0.003}$ | $0.582^{\pm 0.011}$ | $0.713^{\pm 0.003}$ | $0.541^{\pm 0.001}$ |
| Class. | ActProbe | $0.654^{\pm 0.003}$ | $0.558^{\pm 0.006}$ | $0.500^{\pm 0.000}$ | $0.500^{\pm 0.001}$ |
| Class. | Hide-and-Seek | $0.742^{\pm 0.033}$ | $\mathbf{0.599}^{\pm 0.005}$ | $\underline{0.790}^{\pm 0.002}$ | $\mathbf{0.595}^{\pm 0.005}$ |
| Time | Global timer G* | **0.988** | 0.550 | **0.988** | $\underline{0.551}$ |
| Time | Per-task timer P* | $\underline{0.972}$ | $\underline{0.585}$ | – | – |

*Deterministic. P needs the task's own successful rollouts.

TABLE II

WARNING LEAD ON HELD-OUT TEST SETS. EACH MONITOR IS OR-FUSED AT A 5% FPR TARGET WITH P ON SEEN TASKS AND WITH G ON UNSEEN TASKS. $\Delta\bar{\ell}$: GAIN IN MEAN LEAD OVER THAT TIMER. MEAN AND STD OVER 3 SEEDS; **BOLD**/UNDERLINE: BEST/SECOND-BEST MONITOR.

| | **Policy: XR-1 (Success Rate: 55.6%; 380 + 380 test rollouts)** | | | | | | |
|---|---|---|---|---|---|---|---|
| | | Seen tasks + P | | | Unseen tasks + G | | |
| | Method | FPR↓ | $\Delta\bar{\ell}\uparrow$ | TWA↑ | FPR↓ | $\Delta\bar{\ell}\uparrow$ | TWA↑ |
| Time | Timer G | 0.024 | $-0.102$ | 0.550 | 0.024 | – | 0.551 |
| Time | Timer P | 0.055 | – | 0.585 | – | – | – |
| OOD | Mahalanobis | 0.055 | $\underline{0.034}^{\pm 0.000}$ | $\underline{0.602}^{\pm 0.000}$ | 0.073 | $0.044^{\pm 0.001}$ | $0.549^{\pm 0.001}$ |
| OOD | Cosine k-NN | 0.058 | $0.021^{\pm 0.000}$ | $0.594^{\pm 0.000}$ | 0.079 | $\underline{0.053}^{\pm 0.000}$ | $0.550^{\pm 0.000}$ |
| OOD | PCA-KMeans | 0.060 | $0.022^{\pm 0.001}$ | $0.594^{\pm 0.000}$ | 0.071 | $\underline{0.053}^{\pm 0.001}$ | $\underline{0.554}^{\pm 0.000}$ |
| OOD | RND | 0.060 | $0.021^{\pm 0.001}$ | $0.593^{\pm 0.000}$ | 0.073 | $0.050^{\pm 0.001}$ | $0.551^{\pm 0.000}$ |
| Cl. | SAFE-MLP | 0.058 | $0.024^{\pm 0.015}$ | $0.596^{\pm 0.006}$ | 0.053 | $0.028^{\pm 0.009}$ | $0.551^{\pm 0.005}$ |
| Cl. | Hide&Seek | 0.055 | $\mathbf{0.039}^{\pm 0.006}$ | $\mathbf{0.605}^{\pm 0.005}$ | 0.059 | $\mathbf{0.122}^{\pm 0.005}$ | $\mathbf{0.595}^{\pm 0.004}$ |
| Abl. | First step | 0.055 | $0.001^{\pm 0.001}$ | $0.586^{\pm 0.001}$ | 0.027 | $0.002^{\pm 0.003}$ | $0.550^{\pm 0.001}$ |
| Abl. | Zero input | 0.062 | $0.013^{\pm 0.003}$ | $0.588^{\pm 0.000}$ | 0.037 | $0.020^{\pm 0.008}$ | $0.554^{\pm 0.005}$ |
| Abl. | Shuffle | 0.058 | $0.005^{\pm 0.001}$ | $0.586^{\pm 0.002}$ | – | – | – |
| | **Policy: RLDX-1 (Success Rate: 56.2%; 100 + 100 recorded test rollouts)** | | | | | | |
| | | Seen tasks + P | | | Unseen tasks + G | | |
| | Method | FPR↓ | $\Delta\bar{\ell}\uparrow$ | TWA↑ | FPR↓ | $\Delta\bar{\ell}\uparrow$ | TWA↑ |
| Time | Timer G | 0.010 | $-0.239$ | 0.511 | 0.000 | – | 0.513 |
| Time | Timer P | 0.030 | – | 0.621 | – | – | – |
| OOD | Mahalanobis | 0.040 | $0.030^{\pm 0.000}$ | $0.631^{\pm 0.000}$ | 0.030 | $0.112^{\pm 0.000}$ | $0.554^{\pm 0.000}$ |
| OOD | Cosine k-NN | 0.050 | $0.005^{\pm 0.000}$ | $0.613^{\pm 0.000}$ | 0.010 | $0.112^{\pm 0.000}$ | $\underline{0.564}^{\pm 0.000}$ |
| OOD | PCA-KMeans | 0.047 | $0.029^{\pm 0.004}$ | $0.627^{\pm 0.001}$ | 0.040 | $\underline{0.115}^{\pm 0.004}$ | $0.550^{\pm 0.002}$ |
| OOD | RND | 0.040 | $0.015^{\pm 0.007}$ | $0.623^{\pm 0.010}$ | 0.037 | $0.099^{\pm 0.013}$ | $0.544^{\pm 0.011}$ |
| Class. | SAFE-MLP | 0.053 | $0.008^{\pm 0.012}$ | $0.613^{\pm 0.007}$ | 0.070 | $0.044^{\pm 0.000}$ | $0.500^{\pm 0.000}$ |
| Class. | SAFE-LSTM | 0.033 | $\underline{0.036}^{\pm 0.003}$ | $\underline{0.637}^{\pm 0.003}$ | 0.067 | $0.101^{\pm 0.006}$ | $0.530^{\pm 0.000}$ |
| Class. | Hide&Seek | 0.050 | $\mathbf{0.074}^{\pm 0.030}$ | $\mathbf{0.648}^{\pm 0.004}$ | 0.057 | $\mathbf{0.196}^{\pm 0.013}$ | $\mathbf{0.583}^{\pm 0.021}$ |
| Abl. | First step | 0.050 | $-0.004^{\pm 0.014}$ | $0.609^{\pm 0.005}$ | 0.090 | $0.038^{\pm 0.025}$ | $0.487^{\pm 0.012}$ |
| Abl. | Zero input | 0.037 | $0.012^{\pm 0.010}$ | $0.623^{\pm 0.002}$ | 0.023 | $0.010^{\pm 0.024}$ | $0.506^{\pm 0.008}$ |
| Abl. | Shuffle | 0.030 | $0.007^{\pm 0.005}$ | $0.624^{\pm 0.002}$ | – | – | – |

Mean lead of the timers: XR-1 P 0.225, G 0.123 (seen) and 0.126 (unseen); RLDX-1 P 0.271, G 0.032 and 0.026. Unseen FPRs can exceed 0.05, as thresholds are set on the seen tasks.

scores gain at most 0.009, so the gain comes from the policy, not from the fusion itself. At matched false alarms, however, it shrinks to 0.015 ([−0.007, 0.041]).

**Hide-and-Seek: an earlier warning (H1, H2).** Hide-and-Seek + P leads P by 0.071 ([0.022, 0.157]) at the same 0.061 FPR; random and resampled scores gain at most 0.012. At matched false alarms it gains 0.056 ([−0.014, 0.120]), positive

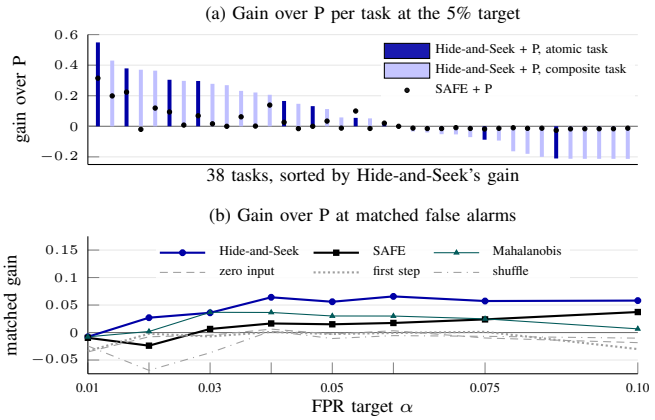



Fig. 2. Where the gain over the per-task timer P comes from (XR-1, held-out test set). (a) Atomic tasks gain about four times as much as composite ones (Hide-and-Seek: 0.160 vs. 0.039 on average); on 16 tasks the fused detector alarms later than P, as its looser timer spends the false-alarm budget on the score. (b) With P loosened to the fused detector's false alarms, Hide-and-Seek's gain stays positive at every target from 2% to 10%; its input ablations never gain more than 0.01.

at every target from 2% to 10%, where SAFE averages 0.013 (Fig. 2b). Its advantage over SAFE + P, 0.041 ([−0.011, 0.103]), is not significant. A single Hide-and-Seek model gains only $0.039 \pm 0.006$ (Table II), so seed averaging is part of its warning.

**Where the gain comes from.** The gain is uneven across tasks (Fig. 2a). All six policy scores gain more on the 10 atomic tasks (0.045 to 0.160) than on the 28 composite ones (0.008 to 0.039). Hide-and-Seek + P alarms earlier than P on 21 tasks and later on 16, where its looser timer margin spends the false-alarm budget on the score; five tasks carry 78% of its gain, and SAFE helps on the same tasks (rank correlation 0.71).

### *D. Is the extra warning learned elapsed time?*

A recurrent detector trained on rollouts that all fail at the limit could learn task durations. The zero-input ablation separates failures as well as Hide-and-Seek (Sec. I), yet adds only 0.001 over P at matched false alarms on the validation set. On the held-out set, Hide-and-Seek beats the first-step, zero-input and shuffle ablations by 0.070, 0.059 and 0.060 (CIs above zero): its warning develops during the rollout and is neither elapsed time nor the task's typical trajectory.

### *E. Unseen tasks, second policy and second benchmark*

**Unseen tasks (Tables I and II).** Known tasks already give the monitors more data than P (48 rollouts per task with failures, against 6 to 45 success durations). With no rollout of the task, as in SAFE's setting [7], only Hide-and-Seek keeps a clear gain over G on XR-1: 0.127 with 3 seeds averaged (zero-input ablation: 0.033), 0.111 more than SAFE ([0.036, 0.166]). SAFE's seed-averaged 0.017 is no larger than the best of 20 random scores (0.019; single models: 0.028), and against G loosened to its 20 false alarms it is −0.079 (Hide-and-Seek: +0.051). The OOD detectors' gains cost three times G's false

TABLE III
FIPER'S BENCHMARK RUN WITH FIPER'S CODE: MEAN LEAD OF FIPER (ITS SELECTED CONFIGURATION, 5 SEEDS) AND OF A TIMER AT FIPER'S TEST FPR.

| Task | FPR | FIPER $\bar{\ell}$ | Timer $\bar{\ell}$ | Difference |
|---|---|---|---|---|
| Pretzel (real) | 0.300 | 0.499 | 0.114 | 0.385 |
| Sorting | 0.430 | 0.386 | 0.215 | 0.170 |
| PushT | 0.558 | 0.664 | 0.586 | 0.079 |
| Stacking | 0.377 | 0.470 | 0.395 | 0.075 |
| PushChair (real)* | 0.064 | 0.348 | 0.329 | 0.019 |
| Average | | | | **0.146** |

*The timer cannot go below 0.20 FPR here, so this difference is a lower bound.

alarms and leave TWA at G's level. Once 1 to 5 successes of the new task exist, a per-task timer can be built, and Hide-and-Seek still adds 0.088 to 0.099 over it (SAFE about 0.01).

**Second policy.** On recorded RLDX-1 rollouts [10] (20 tasks; Table II, bottom), all failures again ran to the time limit. With 3 seeds averaged, SAFE-LSTM + P gains 0.034 ([0.008, 0.067]) and Hide-and-Seek + P, set up with the XR-1 settings before scoring, 0.083 ([0.033, 0.138]; 0.052 at matched false alarms). Both gains come from atomic tasks: on composite tasks no score adds clear lead (Hide-and-Seek 0.010). On unseen tasks, Hide-and-Seek again gains most over G ($0.196 \pm 0.013$; ablations $\leq 0.038$); SAFE-LSTM and the OOD detectors (default settings) gain 0.10 to 0.115 over this late timer (lead 0.026).

**FIPER's benchmark.** In FIPER's released rollouts [23] (five tasks, two on a real robot), length alone separates failures with ROC-AUC 0.92–1.00 in four tasks. We re-ran FIPER's code (reproducing its results within 0.05) with elapsed time as one more method. Under FIPER's thresholds and selection rule, the timer has the best balanced accuracy (0.81 against 0.78) at a test FPR of 0.09, against 0.35 for FIPER. At FIPER's own FPR, however, FIPER warns 0.146 of the episode earlier ([0.104, 0.188]; Table III): the test credits early warning wherever it occurs.

### *F. When do failures become irreversible?*

One annotator, blind to alarms, marked the point of no return (the first state from which the task could not be completed without outside help) on 48 held-out failures: 38 (79%) reached no such state before the limit (*recoverable*), most often because the robot was slow or nearly succeeded; a second annotator agreed on 11 of 12 ($\kappa = 0.80$). An early "failure" alarm thus mostly flags an attempt that is not yet lost. On all 10 failures that did become irreversible, P alarmed after the point of no return, and no detector flags more than 8% of failures by half the limit.

## VII. STAGE 2: PLANNED EXPERIMENTS

**Second policy (E1).** We test H3 on new RLDX-1 rollouts: fresh calibration and test sets of 20 successes and 20 failures for each of its 20 tasks (about 2,500 rollouts). The detectors (SAFE-LSTM, Hide-and-Seek), $H_k$, $\rho_k$ and each fused path are fixed on the existing rollouts, so the new calibration successes set only $\hat{\lambda}$ and Proposition 1 applies to the evaluated detector. Each fused detector is compared with P at the same target and at matched false alarms, and with fusions of random scores.

**Reset and retry (E2).** We test H4 in closed loop on 1,000 RLDX-1 scenes (50 per task) with a budget of $1.25\,H_k$. Each scene runs once without reset; for each monitor, we branch from this run at its first alarm, restore the scene and retry in the remaining budget. We compare the success rates of resets triggered by Hide-and-Seek + P and by P on the same scenes, and report unnecessary resets and wasted time.

**Further analyses (E3).** Action-only and multi-sample monitors, richer time baselines, unseen tasks on new rollouts, and lead measured from the point of no return on 250 to 500 annotated failures.

## VIII. CONCLUSION AND LIMITATIONS

When failure means running out of time, a monitor must beat a timer at the same false-alarm rate. Against a per-task timer at the same target FPR, SAFE added 0.030 of the time limit on a held-out test set and Hide-and-Seek 0.071 (0.015 and 0.056 at matched false alarms), a warning that input ablations trace to the rollout and that carries over to unseen tasks. Stage 2 tests both on a second policy and measures their value in closed-loop retry.

**Limitations.** Alarms were recorded, not acted on, in simulation, and lead is measured to the time limit, not to an irreversible failure. RLDX-1 is evaluated only on recorded rollouts. The timers and fusions were designed on the validation set, and the matched, per-task, unseen-task and FIPER analyses were added after the held-out test set was scored; Hide-and-Seek, the better of two supervised baselines on the validation set, was tested on a held-out set already used for SAFE. Monitors assume a known task and time limit; fixed-horizon data need failure-time labels. Only 10 annotated failures became irreversible.

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